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United States Patent Application Publication

20250282368

Kind Code

A1

Publication Date

September 11, 2025

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CONTROL DEVICE FOR VEHICLE

Abstract

The larger the amount of change in the accelerator operation amount from the point in time when the accelerator returning operation is performed to the current point in time, the larger the reduction rate of the driving torque at the time of decreasing toward the required value of the driving torque is set. When the amount of change in the accelerator operation amount is relatively great, the reduction rate of the driving torque is set to a relatively great value in accordance with the scene in which prioritizing responsiveness is likely to be desired, such that the driving torque is quickly reduced. When the amount of change in the accelerator operation amount is relatively small, the reduction rate of the driving torque is set to a relatively small value in accordance with the scene in which prioritizing controllability is likely to be desired, so the driving torque is gradually reduced.

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Appl. No.: 18/979659

Filed: December 13, 2024

Foreign Application Priority Data

JP 2024-036318

Mar. 08, 2024

Publication Classification

Int. Cl.: B60W50/08 (20200101); B60W10/08 (20060101); B60W10/10 (20120101);
B60W50/10 (20120101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-036318 filed on Mar. 8, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The disclosure relates to a control device for a vehicle that sets a reduction rate of a driving torque when reducing the driving torque toward a requested value.

2. Description of Related Art

[0003] A control device for a vehicle including a power source and a power transmission device that transmits power from the power source to drive wheels is well known. An example thereof is a vehicle described in Japanese Unexamined Patent Application Publication No. 2009-137384 (JP 2009-137384 A). This JP 2009-137384 A discloses that when transitioning from a state in which an accelerator is depressed to a state in which the accelerator is not depressed, there is a tendency for increase in accordance with increase in allowance electric power currently charging a battery up to an input limit, regarding which a reduction rate for driving torque is set so as to be reduced toward a requested driving torque.

SUMMARY

[0004] Now, difference in accelerator operation amount at a point in time of an accelerator returning operation, or difference in accelerator operation amount at a current point in time after the accelerator returning operation, may be difference in accordance with whether the situation is a scene in which prioritizing responsiveness is desired or a scene in which prioritizing controllability is desired. The accelerator returning operation is an accelerator operation state that changes from an increased state to a reduced state. Responsiveness is a deceleration characteristic that rapidly reduces driving torque in response to the accelerator returning operation. Controllability is a deceleration characteristic that gradually reduces the driving torque in response to the accelerator returning operation. Incidentally, in the technology described in JP 2009-137384 A, when the accelerator returning operation is performed, the reduction rate of the driving torque is set in accordance with the input limit of the battery, and a uniform reduction rate of the driving torque is set regardless of the difference in the accelerator operation amount. Accordingly, when the reduction rate of the driving torque is set so as to correspond to a scene in which prioritizing responsiveness is desired, controllability may deteriorate. Alternatively, when the reduction rate of the driving torque is set so as to correspond to a scene in which prioritizing controllability is desired, responsiveness may deteriorate.

[0005] The disclosure has been made in view of the above circumstances, and an object thereof is to provide a control device for a vehicle that is capable of setting a reduction rate of driving torque by differentiating scenes in accordance with responsiveness and controllability.

[0006] An essence of a first aspect of the disclosure is [0007] (a) control device for a vehicle that is equipped with a power source and a power transmission device for transmitting power from the power source to a drive wheel, the control device including [0008] (b) a drive control unit that, the greater an amount of change in an accelerator operation amount from a point in time at which an accelerator returning operation was performed to a current point in time is, sets a reduction rate of driving torque to a greater value, when reducing the driving torque toward a requested value of the driving torque based on a current value of the accelerator operation amount. The accelerator

returning operation is an operation in which an accelerator operation state changes from an increased state to a reduced state.

[0009] According to the first aspect of the disclosure, the greater the amount of change in the accelerator operation amount from the point in time when the accelerator returning operation is performed, to the current point in time, the greater the value to which the reduction rate of the driving torque when being reduced toward the requested value of the driving torque is set to be. As a result, when the amount of change in the accelerator operation amount is relatively great, the reduction rate of the driving torque is set to a relatively great value in accordance with the scene in which prioritizing responsiveness is likely to be desired, such that the driving torque is quickly reduced. On the other hand, when the amount of change in the accelerator operation amount is relatively small, the reduction rate of the driving torque is set to a relatively small value in accordance with the scene in which prioritizing controllability is likely to be desired, such that the driving torque is gradually reduced. Accordingly, the reduction rate of the driving torque can be set by differentiating scenes in accordance with responsiveness and controllability.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0011] FIG. 1 is a diagram for explaining a schematic configuration of a vehicle to which the present disclosure is applied, and is a diagram for explaining a control function for various kinds of control in a vehicle and a main part of a control system;

[0012] FIG. 2 is a flowchart for explaining a main part of a control operation of the electronic control device, and is a flowchart for explaining a control operation for determining whether an accelerator operation state is an ascending state or a descending state;

[0013] FIG. 3 is a diagram for explaining an example of a functional block for setting a torque reduction rate by dividing a scene by responsiveness and controllability;

[0014] FIG. 4 is a flow chart describing a main part of a control operation of the electronic control device, and is a flow chart describing a control operation for setting a torque reduction rate by dividing a scene by responsiveness and controllability; and

[0015] FIG. 5 is a diagram illustrating an example of a time chart when the control operation illustrated in the flowchart of FIG. 4 is executed.

DETAILED DESCRIPTION OF EMBODIMENTS

[0016] Hereinafter, examples of the present disclosure will be described in detail with reference to the drawings.

[0017] FIG. 1 is a diagram illustrating a schematic configuration of a vehicle **10** to which the present disclosure is applied, and is a diagram illustrating control functions and a main part of a control system for various controls in the vehicle **10**. In FIG. 1, the vehicle **10** includes left and right front wheels **12**, a front wheel drive device **20** that drives the front wheels **12**, left and right rear wheels **14**, and a rear wheel drive device **30** that drives the rear wheels **14** separately from each other. Further, the vehicle **10** includes a battery **40** or the like that is a chargeable/dischargeable DC power source. Note that the “left and right” is the left and right with respect to the forward direction of the vehicle **10**.

[0018] The vehicle **10** is an all-wheel drive vehicle capable of adjusting the driving torque distribution of the front wheels **12** and the rear wheels **14**. The vehicle **10** is a vehicle including two wheels each including a front wheel **12** and a rear wheel **14** and including four wheels, and thus is also a four-wheel drive vehicle. In the present embodiment, all-wheel drive (=AWD) and four-

wheel drive (=4WD) have the same meaning. In addition to traveling in 4WD control (4WD state is also synonymous) in which the driving torque is distributed to four wheels, the vehicle **10** can also travel in two-wheel drive (=2WD) control (2WD state is also synonymous) in which the driving torque is distributed only to one of the front wheels **12** and the rear wheels **14**. The front wheels **12** and the rear wheels **14** are drive wheels of the vehicle **10**.

[0019] The front wheel drive device **20** includes a front motor **22**, a front wheel power transmission device **24**, and a front wheel PCU (Power Control Unit) **26**. The rear wheel drive device **30** includes a rear motor **32**, a rear wheel power transmission device **34**, and a rear wheel PCU **36**. The front motor **22** and the rear motor **32** each function as a power source of the vehicle **10**. The front wheel power transmission device **24** is a power transmission device that transmits power from the front motor **22** to the front wheel **12**. The rear wheel power transmission device **34** is a power transmission device that transmits power from the rear motor **32** to the rear wheel **14**. The front wheel PCU **26** controls the front motor **22**. The rear wheel PCU **36** controls the rear motor **32**.

[0020] The front motor **22** is a known rotary electric machine, a so-called motor generator, and is connected to the battery **40** via a front wheel PCU **26**. The front wheel power transmission device **24** includes a reduction gear mechanism including a differential gear (not shown) connected to the front motor **22**, a pair of front drive shafts **28** connected to the reduction gear mechanism, and the like. The front wheel PCU **26** includes, for example, inverters, and converts DC power from the battery **40** into AC power and supplies the AC power to the front motor **22**. Further, the front wheel PCU **26** converts AC power generated by the front motor **22** by, for example, regenerative braking into DC power and supplies the DC power to the battery **40**. The front wheel PCU **26** is a power control device that controls electric power exchanged between the battery **40** and the front motor **22**.

[0021] The rear motor **32**, the rear wheel power transmission device **34**, and the rear wheel PCU **36** each have the same functions as those of the front motor **22**, the front wheel power transmission device **24**, and the front wheel PCU **26**, and their explanations are omitted.

[0022] The vehicle **10** is a vehicle capable of independently controlling the variation of the driving torque of the front and rear wheels, and is a front and rear wheel independent driving type battery electric vehicle, that is, a BEV (Battery Electric Vehicle). In the present embodiment, when the front wheel drive device **20** and the rear wheel drive device **30** are not particularly distinguished from each other, the drive device PU is represented. When the front motor **22** and the rear motor **32** are not particularly distinguished from each other, they are referred to as motor MG. When the front wheel power transmission device **24** and the rear wheel power transmission device **34** are not particularly distinguished from each other, they are referred to as a power transmission device PT.

[0023] The vehicle **10** further includes an electronic control device **50** as a controller including a control device for the vehicle **10** related to various controls such as driving torque control for the front and rear wheels. The electronic control device **50** includes, for example, a so-called microcomputer including a CPU, RAM, ROM, an input/output interface, and the like. CPU performs various kinds of control of the vehicles **10** by performing signal-processing in accordance with a program stored in ROM in advance while using a temporary storage function of RAM.

[0024] Various signals based on detection values by various sensors provided in the vehicle **10** are supplied to the electronic control device **50**. Examples of the various sensors include a MGF rotation sensor **60**, a MGR rotation sensor **62**, wheel speed sensors **64**, an accelerator operation amount sensor **66**, a G sensor **68**, and a yaw rate sensor **70**. The various types of signals include, for example, a MGF rotational speed N_{mgf} , MGR rotational speed N_{mgr} , left and right front wheel rotational speeds N_{wfl} , N_{wfr} , left and right rear wheel rotational speeds N_{wrl} , N_{wrr} , an accelerator operation amount θ_{acc} , a front-rear acceleration G_x , a left-right acceleration G_y , a yaw rate R_{yaw} , and the like. The electronic control device **50** acquires the vehicle speed V based on the front wheel rotational speed N_{wfl} , N_{wfr} and the rear wheel rotational speed N_{wrl} , N_{wrr} . For example, the electronic control device **50** acquires the vehicle speed V from the average value of the front wheel

rotational speed N_{wfl} , N_{wfr} and the rear wheel rotational speed N_{wrl} , N_{wrr} , the average value of the front wheel rotational speed N_{wfl} , N_{wfr} , the average value of the rear wheel rotational speed N_{wrl} , N_{wrr} , or the like.

[0025] MGF rotational speed N_{mgf} is the rotational speed of the front motor **22**. MGR rotational speed N_{mgr} is the rotational speed of the rear motor **32**. The front wheel rotational speed N_{wfl} , N_{wfr} is the rotational speed of the front wheel **12**. The rear wheel rotational speed N_{wrl} , N_{wrr} is the rotational speed of the rear wheel **14**. The accelerator operation amount θ_{acc} is a signal indicating a magnitude of an acceleration operation of the driver (=driver), and is an accelerator operation amount of the driver. The front-rear acceleration G_x is a longitudinal acceleration of the vehicle **10**. The left-right acceleration G_y is an acceleration of the vehicle **10** in the left-right direction. The yaw rate R_{yaw} is a rotational angular velocity around the vertical axis of the vehicle **10**.

[0026] Various command signals are output from the electronic control device **50** to each device provided in the vehicle **10**. The devices are, for example, a front wheel PCU **26**, a rear wheel PCU **36**, and the like. The various command signals are, for example, a MGF control command signal Sm_{gf} , MGR control command signal Sm_{gr} . MGF control command Sm_{gf} is a torque command for controlling MGF torque T_{mgf} which is the torque of the front motor **22**. MGR control command Sm_{gr} is a torque command for controlling MGR torque T_{mgr} which is the torque of the rear motor **32**.

[0027] The electronic control device **50** includes a drive control unit **52** for realizing various kinds of control in the vehicle **10**.

[0028] The drive control unit **52** controls the drive device PU. For example, the drive control unit **52** calculates a drive request amount for the vehicle **10** by the driver by applying the accelerator operation amount θ_{acc} and the vehicle speed V to the drive request amount map. The drive request amount map is a relationship for determining a drive request amount that is determined and stored in advance experimentally or by design, that is, predetermined. For example, a required driving torque T_{wdem} as a requested value of a driving torque T_w which is a torque in the wheels (the front wheels **12** and the rear wheels **14**) is used as the drive required amount. The drive control unit **52** performs drive control for controlling the motor MG so as to realize the required driving torque T_{wdem} in view of the transmission efficiency of the power transmission device PT and the like. The torque and the force (driving force) are synonymous with each other unless otherwise distinguished. The accelerator operation amount θ_{acc} is synonymous with the current accelerator operation amount θ_{accr} , which is the current value of the accelerator operation amount θ_{acc} , unless otherwise specified.

[0029] The driving torque T_w is a combination of the front torque T_{wf} which is the torque in the front wheel **12** and the rear torque T_{wr} which is the torque in the rear wheel **14**. The required driving torque T_{wdem} is a value obtained by combining the required front torque T_{wfdem} as the requested value of the front torque T_{wf} and the required rear torque T_{wrdem} as the requested value of the rear torque T_{wr} . The drive control unit **52** sets the torque distribution ratio γ_{fr} of the front and rear wheels by applying a plurality of driving force-related values such as the accelerator operation amount θ_{acc} , the vehicle speed V , the front-rear acceleration G_x , and the yaw rate R_{yaw} to the torque distribution ratio map, for example. The torque distribution ratio map is a relationship for obtaining a torque distribution ratio γ_{fr} based on a predetermined traveling state. The drive control unit **52** calculates the required front torque T_{wfdem} and the required rear torque T_{wrdem} based on the required driving torque T_{wdem} and the torque distribution ratio γ_{fr} . The drive control unit **52** outputs a MGF control command Sm_{gf} for controlling the front motor **22** so as to realize the required front torque T_{wfdem} to the front wheel PCU **26**. The drive control unit **52** outputs a MGR control command Sm_{gr} for controlling the rear motor **32** so as to realize the required rear torque T_{wrdem} to the rear wheel PCU **36**.

[0030] Here, when an accelerator returning operation in which the accelerator operation state

changes from the increase state to the decrease state is performed by the driver, the drive control unit 52 sets a torque reduction rate R_{dt} for decreasing the accelerator operation state toward the required driving torque T_{wdem} . The torque reduction rate R_{dt} is a reduction rate of the driving torque T_w . When the accelerator returning operation is performed, the drive control unit 52 reduces the driving torque T_w by the torque reduction rate R_{dt} . The drive control unit 52 sets, for the required front torque T_{wfdem} , a torque reduction rate R_{dt} on the front wheel 12 side (the front side is also synonymous), that is, a front torque reduction rate R_{dtf} which is a reduction rate of the front torque T_{wf} . The drive control unit 52 sets, for the required rear torque T_{wrdem} , a rear torque reduction rate R_{dtr} which is a torque reduction rate R_{dt} on the rear wheel 14 side (also synonymous with the rear side), that is, a reduction rate of the rear torque T_{wr} . When the accelerator returning operation is performed, the drive control unit 52 reduces the front torque T_{wf} by the front torque reduction rate R_{dtf} and reduces the rear torque T_{wr} by the rear torque reduction rate R_{dtr} .

Increasing the accelerator operation state is a state in which the accelerator operation amount θ_{acc} is increased or a state in which the accelerator operation amount θ_{acc} is maintained at a value after the increase or in the vicinity of the value. The reduced state of the accelerator operation state is a state in which the accelerator operation amount θ_{acc} is lowered or a state in which the accelerator operation amount θ_{acc} is maintained at a value after the lowering or in the vicinity of the value. In the accelerator operation state and the accelerator operation amount θ_{acc} , the increase and the increase are synonymous with each other, and the decrease and the decrease are synonymous with each other. The torque reduction rate R_{dt} is expressed by an absolute value for convenience, and the larger the value, the more the driving torque T_w tends to decrease.

[0031] Incidentally, for example, when the torque reduction rate R_{dt} is set according to the present accelerator operation amount θ_{accr} , the torque reduction rate R_{dt} corresponding to the accelerator operation amount θ_{acc} prior to the accelerator returning operation is not set. For example, the torque reduction rate R_{dt} is set to be the same when the accelerator operation amount θ_{acc} is from the high opening degree and when the accelerator operation amount θ_{acc} is from the low opening degree. On the other hand, a scene of an accelerator returning operation in which the accelerator operation amount θ_{acc} is changed from a high opening degree to a steady travel equivalent is a scene in which responsiveness is desired to be emphasized. Alternatively, a scene of the accelerator returning operation in which the accelerator operation amount θ_{acc} is changed from the low opening degree to the steady travel equivalent is a scene in which the controllability is desired to be emphasized. When the torque reduction rate R_{dt} is set uniformly in accordance with the accelerator operation amount θ_{accr} at present, it is not possible to separate a scene in which responsiveness is desired to be emphasized and a scene in which controllability is desired to be emphasized.

Therefore, when the torque reduction rate R_{dt} is set on the basis of the accelerator returning operation in which the accelerator operation amount θ_{acc} is changed from the high opening degree to the steady travel equivalent, a relatively large torque reduction rate R_{dt} is set regardless of the accelerator operation amount θ_{acc} prior to the accelerator returning operation. In this case, when an accelerator returning operation is performed in which the accelerator operation amount θ_{acc} is changed from the low opening degree to the steady travel equivalent, the controllability may deteriorate. Alternatively, when the torque reduction rate R_{dt} is set on the basis of the accelerator returning operation in which the accelerator operation amount θ_{acc} is changed from the low opening degree to the steady travel equivalent, a relatively small torque reduction rate R_{dt} is set regardless of the accelerator operation amount θ_{acc} prior to the accelerator returning operation. In this case, there is a possibility that the responsiveness deteriorates when an accelerator returning operation is performed in which the accelerator operation amount θ_{acc} is changed from the high opening degree to the steady travel equivalent.

[0032] The above-described problem that it is impossible to separate a scene in which responsiveness is desired to be emphasized from a scene in which controllability is desired to be emphasized may occur even when the torque reduction rate R_{dt} is uniformly set in accordance with

the accelerator operation amount θ_{acc} prior to the accelerator returning operation. For example, the above-described problem may occur even when the torque reduction rate R_{dt} is set to the same value in a case where the accelerator returning operation performed from the accelerator operation amount θ_{acc} having the same value is the current accelerator operation amount θ_{accr} to the high opening degree and the low opening degree.

[0033] Therefore, the drive control unit **52** sets the torque reduction rate R_{dt} at the time of decreasing toward the required driving torque T_{wdem} to a larger value as the variation of the accelerator operation amount θ_{acc} from the point in time when the accelerator returning operation is performed to the current point in time is larger. The amount of change in the accelerator operation amount θ_{acc} from the point in time when the accelerator returning operation is performed to the current point in time is the difference between the accelerator operation amount θ_{acc} at the point in time when the accelerator returning operation is performed and the current accelerator operation amount θ_{accr} .

[0034] For example, the drive control unit **52** sets the torque reduction rate R_{dt} to a larger value as compared with a case where the accelerator operation amount θ_{acc} is made from a smaller value when the accelerator operation amount θ_{acc} is made from a larger value when the accelerator returning operation to the present accelerator operation amount θ_{accr} of the same value is made from a larger value. And/or, when the accelerator returning operation performed from the accelerator operation amount θ_{acc} having the same value is the accelerator operation amount θ_{acc} having the same value and the accelerator operation amount θ_{accr} having the smaller value, the drive control unit **52** sets the torque reduction rate R_{dt} to a larger value as compared with the case of the accelerator operation amount θ_{accr} having the larger value.

[0035] The drive control unit **52** includes, for example, an accelerator operation state determination unit **54**, a reference operation amount setting unit **56**, and a torque reduction rate setting unit **58** in order to set a torque reduction rate R_{dt} by dividing a scene by responsiveness and controllability.

[0036] FIG. **2** is a flowchart for explaining a main part of the control operation of the electronic control device **50**, and is a flowchart for explaining a control operation for determining whether the accelerator operation state is an ascending state or a descending state, and is repeatedly executed, for example.

[0037] In FIG. **2**, each step of the flowchart corresponds to the function of the accelerator operation state determination unit **54**. In **S10** of the step (hereinafter, the step is omitted), it is determined whether or not the accelerator operation amount θ_{accr} is larger than the value obtained by adding the predetermined increase opening degree to the previous value of the accelerator operation amount θ_{acc} . The previous value of the accelerator operation amount θ_{acc} is, for example, a value before a predetermined cycle or a value before a predetermined time in the flowchart. The predetermined ascending opening degree is, for example, a predetermined hysteresis equivalent amount for determining that the accelerator operation amount θ_{acc} is ascending. When the determination of **S10** is affirmative, it is determined in **S20** that the accelerator operating condition is elevated. If the determination of **S10** is negative, it is determined in **S30** whether or not the present accelerator operation amount θ_{accr} is smaller than a value obtained by subtracting the predetermined lowering opening degree from the previous value of the accelerator operation amount θ_{acc} . The predetermined descent opening degree is, for example, a predetermined hysteresis equivalent amount for determining that the accelerator operation amount θ_{acc} is descending. When the determination of **S30** is affirmative, it is determined in **S40** that the accelerator operating condition is lowered. When the determination of **S30** is negative, the previous value of the ascent determination or the descent determination is held in **S50**. This previous value is a value one cycle before the flowchart.

[0038] FIG. **3** is a diagram for explaining an exemplary functional block for setting a torque-reduction-rate R_{dt} by dividing a scene by responsiveness and controllability. In FIG. **3**, the blocking **B10** corresponds to the function of the accelerator operation state determination unit **54**.

The blocking **B20** corresponds to the function of the reference operation amount setting unit **56**.

The blocking **B30** corresponds to the function of the torque reduction rate setting unit **58**.

[0039] In the blocking **B10**, the accelerator operating condition is determined based on the present accelerator operation amount θ_{accr} . For example, based on the current accelerator operation amount θ_{accr} , it is determined whether the accelerator operation state is an increase or a decrease (see FIG. 2). In addition, it is determined whether or not an accelerator returning operation has been performed based on whether or not the accelerator operation state has changed from ascending to descending. In this way, the accelerator operation state determination unit **54** determines whether or not an accelerator returning operation has been performed.

[0040] In the blocking **B20**, the accelerator-off reference opening degree θ_{accb} is calculated based on the accelerator operation amount θ_{accr} and the accelerator operating condition. The accelerator off reference opening degree θ_{accb} is an accelerator operation amount θ_{acc} that is a reference when the amount of change in the accelerator operation amount θ_{acc} from the point in time when the accelerator returning operation is performed to the current point in time is obtained, that is, a reference operation amount. For example, when the determination that the accelerator returning operation has been performed is not input, the current accelerator-off reference opening degree θ_{accb} is held. When the determination that the accelerator returning operation has been performed is input, the accelerator-off reference opening degree θ_{accb} is updated to the value of the current accelerator operation amount θ_{accr} at that time. As described above, the reference operation amount setting unit **56** sets the accelerator operation amount θ_{acc} at the point in time when the accelerator returning operation is performed as the accelerator off reference opening degree θ_{accb} .

[0041] In the blocking **B30**, the torque-reduction-rate R_{dt} is calculated based on the accelerator operation amount θ_{accr} and the accelerator-off reference opening θ_{accb} . The front torque reduction rate R_{dtf} and the rear torque reduction rate R_{dtr} are respectively calculated so that the driving torque T_w is reduced by the calculated torque reduction rate R_{dt} . For example, the torque reduction rate R_{dt} is set by applying the present accelerator operation amount θ_{accr} and the accelerator-off reference opening degree θ_{accb} to the torque reduction rate map. The torque reduction rate map is a relation for determining a torque reduction rate R_{dt} based on a predetermined accelerator operating condition. In the torque reduction rate map, for example, a larger torque reduction rate R_{dt} is set as the reference opening degree difference $\Delta\theta_{accb}$ ($=\theta_{accb}-\theta_{accr}$), which is a difference between the accelerator-off reference opening degree θ_{accb} and the current accelerator operation amount θ_{accr} , is larger. For example, when the present accelerator operation amount θ_{accr} is the same value, a torque reduction rate R_{dt} having a larger value is set as compared with a value in which the accelerator-off reference opening degree θ_{accb} is larger. When the accelerator-off reference opening degree θ_{accb} is the same value, a torque reduction rate R_{dt} having a larger value is set as compared with a value in which the accelerator operation amount θ_{accr} is smaller.

[0042] As described above, the torque reduction rate setting unit **58** sets the torque reduction rate R_{dt} to a larger value as the reference opening degree difference $\Delta\theta_{accb}$ is larger. For example, the torque reduction rate setting unit **58** sets the torque reduction rate R_{dt} to a larger value when the accelerator-off reference opening degree θ_{accb} is a larger value than the current accelerator operation amount θ_{accr} when the accelerator-off reference opening degree θ_{accb} is a larger value than when the accelerator-off reference opening degree θ_{accb} is a smaller value. When the current accelerator operation amount θ_{accr} is a smaller value than the accelerator off reference opening degree θ_{accb} , the torque reduction rate setting unit **58** sets the torque reduction rate R_{dt} to a larger value than when the current accelerator operation amount θ_{accr} is a larger value.

[0043] FIG. 4 is a flow chart for describing a main part of the control operation of the electronic control device **50**, and is a flow chart for describing a control operation for setting a torque-reduction-rate R_{dt} by dividing a scene by responsiveness and controllability, and is repeatedly executed, for example.

[0044] In FIG. 4, first, in **S110** corresponding to the function of the accelerator operation state

determination unit **54**, it is determined whether or not the accelerator operation state has changed from ascending to descending. When the determination of **S110** is affirmative, the accelerator-off reference opening degree θ_{accb} is updated to the present accelerator operation amount θ_{accr} in **S120** corresponding to the function of the reference operation amount setting unit **56**. When the determination of **S110** is negative, the previous value of the accelerator-off reference opening degree θ_{accb} is held in **S130** corresponding to the function of the reference operation amount setting unit **56**. This previous value is a value one cycle before the flowchart. After **S120** or **S130**, the torque reduction rate Rdt is calculated based on the current accelerator operation amount θ_{accr} and the accelerator-off reference opening degree θ_{accb} in **S140** corresponding to the function of the torque reduction rate setting unit **58**. **S110** corresponds to the blocking **B10** of FIG. 3. **S120** and **S130** correspond to the blocking **B20** of FIG. 3. **S140** corresponds to the blocking **B30** of FIG. 3.

[0045] FIG. 5 is a diagram illustrating an example of a time chart when the control operation illustrated in the flowchart of FIG. 4 is executed. In FIG. 5, t_1 point in time indicates a point in time at which it is determined that the accelerator operation state has changed from ascending to descending due to the accelerator being turned off from the accelerator fully opened state, that is, a point in time at which it is determined that the accelerator returning operation has been performed. Then, the present accelerator operation amount θ_{accr} at t_1 point in time is set to the accelerator-off reference opening degree θ_{accb} . As the accelerator operation amount θ_{accr} is reduced from t_1 point in time, the reference opening degree difference $\Delta\theta_{accb}$ is increased, and the front torque reduction rate Rdt_f and the rear torque reduction rate Rdt_r are respectively increased (refer to t_1 point in time to t_2 point in time). When the accelerator is turned on at t_2 point in time, the accelerator operating condition is changed to an elevated state (refer to t_2 point in time to t_3 point in time). After that, it is determined that the accelerator is turned off again at t_3 point in time, and thus the accelerator returning operation is performed, and the accelerator-off reference opening degree θ_{accb} is updated to the present accelerator operation amount θ_{accr} at t_3 point in time. Thereafter, similarly, at t_4 point in time, the accelerator operation status is changed to an elevated state, and at t_5 point in time, it is determined that the accelerator returning operation has been performed, and the accelerator-off reference opening degree θ_{accb} is updated to the present accelerator operation amount θ_{accr} . For convenience, the front torque reduction rate Rdt_f and the rear torque reduction rate Rdt_r are also calculated while the accelerator operating condition is determined to be increased. However, the torque reduction rate Rdt is not reflected in the drive control of the vehicle **10** during the period in which the accelerator operating state is determined to be increased.

[0046] As described above, according to the present embodiment, as the variation of the accelerator operation amount θ_{acc} from the point in time when the accelerator returning operation is performed to the present point in time is larger, the torque reduction rate Rdt when the accelerator operation amount acc is decreased toward the required driving torque T_{wdem} is set to be larger. As a result, when the variation of the accelerator operation amount θ_{acc} is relatively large, the torque reduction rate Rdt is set to a relatively large value in contrast to the situation in which responsiveness is desired to be emphasized, and thus the driving torque T_w is quickly reduced. On the other hand, when the amount of change in the accelerator operation amount θ_{acc} is relatively small, the torque reduction rate Rdt is set to a relatively small value in contrast to a situation in which controllability is desired to be emphasized, so that the driving torque T_w is gradually reduced. Therefore, the torque reduction rate Rdt can be set by dividing the scene into responsiveness and controllability.

[0047] Further, according to the present embodiment, it is determined whether or not an accelerator returning operation has been performed. The accelerator operation amount θ_{acc} at the point in time when the accelerator returning operation is performed is set as the accelerator-off reference opening degree θ_{accb} . The larger the reference opening degree difference $\Delta\theta_{accb}$ ($=\theta_{accb}-\theta_{accr}$) is, the larger the torque reduction rate Rdt is set. Accordingly, the torque-reduction-rate Rdt can be appropriately set by dividing the scene into responsiveness and controllability.

[0048] Further, according to the present embodiment, when the accelerator-off reference opening

degree θ_{accb} is a large value with respect to the current accelerator operation amount θ_{accr} , the torque reduction rate R_{dt} is set to a larger value than when the accelerator-off reference opening degree θ_{accb} is a small value. Accordingly, when the reference opening degree difference $\Delta\theta_{accb}$ is relatively large, the torque reduction rate R_{dt} is appropriately set to a relatively large value. On the other hand, when the reference opening degree difference $\Delta\theta_{accb}$ is relatively small, the torque reduction rate R_{dt} is appropriately set to a relatively small value.

[0049] Further, according to the present embodiment, when the current accelerator operation amount θ_{accr} is a smaller value than the accelerator off reference opening degree θ_{accb} , the torque reduction rate R_{dt} is set to a larger value than when the current accelerator operation amount θ_{accr} is a larger value. Accordingly, when the reference opening degree difference $\Delta\theta_{accb}$ is relatively large, the torque reduction rate R_{dt} is appropriately set to a relatively large value. On the other hand, when the reference opening degree difference $\Delta\theta_{accb}$ is relatively small, the torque reduction rate R_{dt} is appropriately set to a relatively small value.

[0050] Further, according to the present embodiment, when the accelerator returning operation to the present accelerator operation amount θ_{accr} having the same value is performed from a value having a large accelerator operation amount θ_{acc} , the torque reduction rate R_{dt} is set to a value larger than that in the case where the accelerator operation amount θ_{acc} is performed from a value having a small accelerator operation amount θ_{acc} . And/or when the accelerator returning operation made from the accelerator operation amount θ_{acc} having the same value is to a value with the current accelerator operation amount θ_{accr} being smaller, the torque reduction rate R_{dt} is set to a larger value than when the current accelerator operation amount θ_{accr} is to a larger value. Accordingly, when the reference opening degree difference $\Delta\theta_{accb}$ is relatively large, the torque reduction rate R_{dt} is appropriately set to a relatively large value. On the other hand, when the reference opening degree difference $\Delta\theta_{accb}$ is relatively small, the torque reduction rate R_{dt} is appropriately set to a relatively small value.

[0051] Although the examples of the present disclosure have been described in detail with reference to the drawings, the present disclosure also applies to other modes.

[0052] For example, in the above-described embodiment, BEV of the front and rear wheel independent drive type is exemplified as the vehicles **10**, but the present disclosure is not limited to this embodiment. For example, the vehicles **10** may be electrified vehicle such as known HEV (Hybrid Electric Vehicle), known PHEV (Plug-in Hybrid Electric Vehicle), known FCEV (Fuel Cell Electric Vehicle). The vehicle **10** may be a known engine vehicle. Alternatively, the vehicle **10** may be a known AWD vehicle in which power is distributed to front and rear wheels by transfer or the like, or may be a known 2WD vehicle. In short, the present disclosure can be applied to a vehicle including a power source and a power transmission device that transmits power from the power source to drive wheels.

[0053] It should be noted that the embodiment described above is merely one embodiment, and the present disclosure can be implemented in a mode in which various changes and improvements are made based on the knowledge of those skilled in the art.

Claims

1. A control device for a vehicle that is equipped with a power source and a power transmission device for transmitting power from the power source to a drive wheel, the control device comprising a drive control unit that, the greater an amount of change in an accelerator operation amount from a point in time at which an accelerator returning operation was performed to a current point in time is, sets a reduction rate of driving torque to a greater value, when reducing the driving torque toward a requested value of the driving torque based on a current value of the accelerator operation amount, wherein the accelerator returning operation is an operation in which an accelerator operation state changes from an increased state to a reduced state.

2. The control device according to claim 1, wherein the drive control unit includes an accelerator operation state determination unit for determining whether the accelerator returning operation has been performed, a reference operation amount setting unit for setting the accelerator operation amount, at the point in time at which the accelerator returning operation is performed, as a reference operation amount, and a torque reduction rate setting unit that sets the reduction rate of the driving torque to a greater value the greater a difference between the reference operation amount and the current value of the accelerator operation amount is.
 3. The control device according to claim 2, wherein the torque reduction rate setting unit sets the reduction rate of the driving torque to a greater value when the reference operation amount is a greater value with respect to a current value of the accelerator operation amount, as compared to when the reference operation amount is a smaller value.
 4. A control device according to claim 2, wherein the torque reduction rate setting unit sets the reduction rate of the driving torque to a greater value when a current value of the accelerator operation amount is a smaller value with respect to the reference operation amount, as compared to when a current value of the accelerator operation amount is a greater value.
 5. The control device according to claim 1, wherein the drive control unit sets the reduction rate of the driving torque to a greater value when the accelerator returning operation to a current value of the accelerator operation amount is performed from a value of which the accelerator operation amount is great, as compared to when performed from a value of which the accelerator operation amount is small, in which the accelerator operation amount is the same value for both, or sets the reduction rate of the driving torque to a greater value when the accelerator returning operation is performed to a smaller value of a current value of the accelerator operation amount, as compared to when performed to a greater value of a current value of the accelerator operation amount, in which the accelerator operation amount is the same value for both.
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